

RL-ACID: Reinforcement Learning-Optimized Adaptive Causal Discovery for Robust Anomaly Detection in Industrial Systems

Hechen Yang¹, Yu Yao¹, *Member, IEEE*, Licheng Yang¹, and Wei Yang¹

Abstract—Anomaly detection is essential for the security of industrial control systems. However, dynamic operating conditions introduce nonstationarity and time-varying causal structures, degrading performance and undermining interpretability. To address this, we propose RL-ACID, a lightweight reinforcement learning framework for adaptive causal discovery. It reformulates anomaly detection as sequential causal discovery, introducing the first unified architecture that integrates lightweight reinforcement learning-based search with causal clustering to resolve the adaptability, efficiency, and interpretability tradeoff. Our framework employs a joint time-frequency encoder to extract and construct candidate causal graphs. Building upon this, we design the causal reinforcement learning-based lightweight search algorithm, which formulates graph exploration as a sequential decision process under sparsity and acyclicity constraints, enabling iterative causal structure optimization. To further enhance adaptability in dynamic environments, we introduce a causal clustering module that softly assigns time-varying graphs to latent operational modes through structural experts, thereby distinguishing normal operational fluctuations from true anomalies. Extensive experiments on multiple industrial benchmarks demonstrate the superior performance of RL-ACID. Our framework not only achieves higher accuracy than baselines but also provides interpretable anomaly analysis through causal path tracing.

Index Terms—Anomaly detection, causal discovery, industrial control system (ICS), reinforcement learning (RL).

I. INTRODUCTION

INDUSTRIAL control systems (ICS) are an essential component of modern infrastructure. They facilitate the automated control and monitoring of intricate physical processes in vital sectors, such as energy, manufacturing, and transportation. However, as ICS environments grow in scale and complexity and integrate more deeply with information networks, new challenges

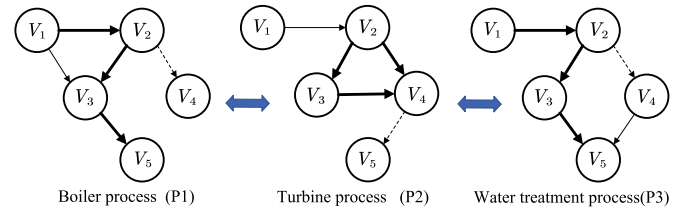


Fig. 1. Causal structure evolution of sensors V_1 – V_5 across operational conditions (HIL-HAI dataset, PCMCi+ algorithm). Edge thickness denotes causal strength. (V_1 : Boiler flow; V_2 : Pressure; V_3 : Valve; V_4 : Vibration; V_5 : Water flow).

to system reliability and security emerge. In ICS, anomalies and faults may originate from equipment degradation, malicious attacks, or operational mistakes. These disturbances have the potential to rapidly propagate across interdependent components, with the consequence of significant system failures [1].

In order to ensure the stable operation of ICS, it is essential to accurately identify and interpret abnormalities, which have become pivotal tasks. In recent years, data-driven methods have made significant progress in anomaly detection, primarily based on machine learning and deep learning models, which show better detection performance [2]. However, such approaches still face significant challenges in industrial real-world scenarios. The majority of models are predicated on statistical correlation between system variables, with the underlying causal relationships being ignored. This reduces interpretability and obstructs root cause analysis.

Furthermore, it is noteworthy that varying operating conditions in ICS often lead to dynamic changes in the underlying causal structure [3]. We investigated this with the hardware-in-the-loop based augmented industrial control system security (HAI) dataset [4], which contains data from a boiler, a turbine, and a water treatment system. We applied the Peter–Clark momentary conditional independence plus (PCMCi+) [5] algorithm to perform causal discovery analysis, focusing on five representative sensors (V_1 to V_5) across three operational modes to illustrate causal drift. As shown in Fig. 1, both the causal connections and their strength, indicated by edge thickness, changed substantially across different operating modes. These changes reflect system-wide adaptive behavior. Our observations were consistent with sensor descriptions documented in the HAI literature, confirming that causal drift is a common rather than an isolated phenomenon.

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Hechen Yang, Yu Yao, and Licheng Yang are with the College of Computer Science and Engineering, Northeastern University, Shenyang 110169, China (e-mail: 2490267@stu.neu.edu.cn; yaoyu@mail.neu.edu.cn; 2310749@stu.neu.edu.cn).

Wei Yang is with the College of Software, Northeastern University, Shenyang 110169, China (e-mail: yangwei@mail.neu.edu.cn).

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Most current anomaly detection models assume that training and monitoring data are independent and identically distributed [6]. However, real ICS exhibit nonstationary behavior due to their complex structure and variable load conditions. This distribution drift compromises model robustness and generalization. Therefore, it is essential to develop adaptive models that can track causal changes over time to enable reliable anomaly detection in dynamic ICS environments.

Current anomaly detection approaches fall into statistical and machine learning categories [7]. Traditional statistical methods identify anomalies by measuring deviations from normal baselines using standard deviation, distance, or density metrics. Techniques like wavelet analysis perform well in periodic simulations but struggle with the high-dimensional, nonstationary nature of ICS data, making it difficult to capture the complex sensor interactions inherent to such environments.

Deep learning advances ICS anomaly detection through neural networks and convolutional architectures that capture complex temporal patterns [8]. However, these models learn statistical correlations rather than causal mechanisms, limiting interpretability and constraining the ability to explain anomaly generation and propagation.

Graph-based methods leverage graph neural networks to capture interdependencies among sensors [9]. Recent advances have introduced dynamic line graphs [10] and federated contrastive frameworks [11] to enhance intrusion detection in network systems, focusing on topological interactions. These approaches are effective at modeling topological interactions for network intrusion detection. However, in dynamic industrial control environments where causal relationships evolve with operational conditions, a key challenge remains in adaptively discovering the underlying time-varying graph structures from data, rather than relying on predefined or static assumptions.

Causal discovery methods have gained attention for enhancing interpretability in anomaly detection. While traditional constraint-based [12] and score-based approaches like the PCMCi+ algorithm infer causality from correlations under strong assumptions, they typically presume static causal structures—limiting their applicability in dynamic industrial environments. Recent work by Chen et al. [13] introduced a root cause identification method using causal graphs and maximum spanning trees, while Qiu et al. [14] proposed a knowledge graph and causal mining approach for root cause analysis. However, these methods often rely on static causal models, which may miss evolving relationships and lead to undetected anomalies. Furthermore, within ICS, large-scale combinatorial searches and repeated statistical evaluations on high-dimensional data result in high computational costs.

In recent years, reinforcement learning (RL) has emerged as a promising approach for dynamic environments due to its inherent capacity for sequential decision-making and adaptation [15]. This makes it particularly well-suited for scenarios where the underlying system properties evolve over time [16]. However, the RL methods often focus on myopic optimization at each step, which demands substantial computational resources through iterative feedback and can hinder their ability to track long-term structural dynamics. Consequently, their

effectiveness in resource-constrained systems like ICS, especially for sustained anomaly detection tasks, remains limited.

Based on the above analysis, we identify three major challenges in current research: *M1: Causal drift*: Conventional static models and anomaly detection methods cannot adapt to dynamic industrial settings. Time-varying causal relationships among sensors reduce model robustness and lead to performance degradation in complex ICS. *M2: High computational cost*: Methods such as causal discovery and RL require substantial computational resources. This makes them difficult to deploy in practical industrial anomaly detection scenarios where efficiency is critical. *M3: Lack of interpretability*: Most models rely on statistical correlations, which fail to capture the underlying causal mechanisms of anomalies. This limitation hinders the understanding of anomaly propagation and root cause analysis in complex systems.

To overcome these limitations, we propose reinforcement learning-optimized adaptive causal discovery for robust anomaly detection (RL-ACID), a lightweight and adaptive causal discovery framework for industrial systems. As illustrated in Fig. 2, our framework comprises three key components. First, we design a sparsity-aware time-frequency transformer encoder to capture both transient dynamics and long-range periodic dependencies in sensor data. This encoder generates compact multiscale representations, which are then processed by a probabilistic decoder to produce an initial weighted adjacency matrix. This matrix serves as a data-informed prior for subsequent causal search (see Section III-A).

Second, we introduce the causal reinforcement learning-based lightweight (CaRLite) algorithm, a lightweight RL-based causal search mechanism. It formulates graph exploration as a sequential decision process, guided by a reward function that promotes data fidelity, acyclicity, and sparsity. Through hierarchical experience replay and causal drift detection, CaRLite efficiently mitigates causal drift (M1) while maintaining high computational efficiency (M3) (see Section III-B).

Finally, to further handle time-varying dependencies, we design a causal clustering module that monitors structural drift by maintaining a set of representative causal graphs learned across different operating modes. A causal inference-based anomaly detection module quantifies deviations from expected behavior and traces causal paths, offering interpretable insights into root causes (M3) (see Section III-C).

The main contributions of this article are as follows.

1) *Novel integration paradigm*: We propose RL-ACID, the first framework that integrates RL-based causal discovery with sparse attention for industrial anomaly detection, enabling synergistic adaptation between temporal patterns and structural constraints.

2) *Lightweight causal discovery*: We design CaRLite, a constrained RL algorithm that achieves 68% efficiency gains through intelligent experience management, resolving the complexity-deployability tradeoff in causal learning.

3) *Dynamic adaptation*: We develop a causal clustering module that autonomously distinguishes structural drift from normal fluctuations, enabling continuous adaptation to evolving industrial processes.

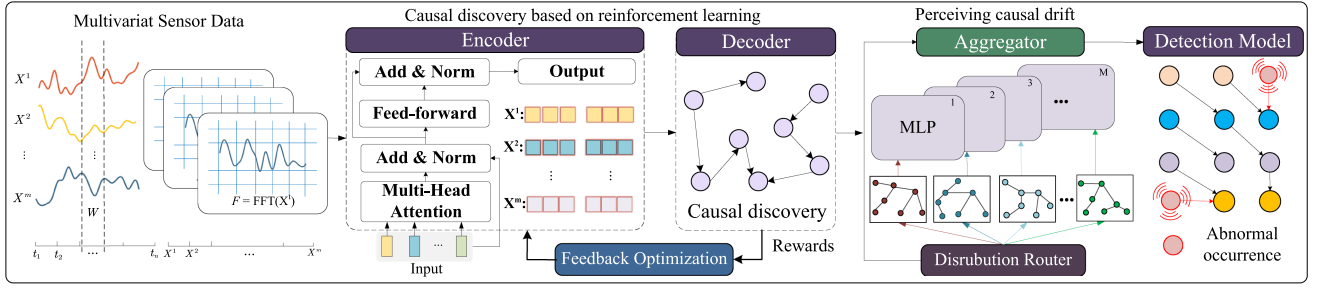


Fig. 2. Architecture of the RL-ACID model.

4) *Theoretical-practical validation*: Extensive experiments demonstrate RL-ACID's comprehensive superiority in detection accuracy, interpretability, and computational efficiency, confirming robust performance under dynamic ICS conditions.

II. PROBLEM STATEMENT, MOTIVATION, AND NOVELTY

A. Problem Formulation

Dynamic industrial environments present three fundamental challenges for reliable anomaly detection: causal drift (M1), computational constraints (M2), and interpretability demands (M3). As illustrated in the example of Fig. 1, where sensor relationships evolve with operational modes, these challenges necessitate reformulation anomaly detection as a time-varying causal discovery and adaptive learning problem.

Formally, we consider a nonstationary multivariate time series $\mathbf{X} \in \mathbb{R}^{T \times D}$, collected from an ICS with D sensors over T time steps. This time series exhibits inherent nonstationarity due to dynamic operating conditions. The objective of this work is twofold. First, we aim to learn a time-varying causal graph $\mathcal{G}_t = (\mathcal{V}, \mathcal{E}_t)$ that captures dynamic dependencies among sensors, where $\mathcal{V}$ denotes the set of sensor nodes and $\mathcal{E}_t$ represents the time-evolving causal edges. Second, we seek to derive a lightweight and interpretable detection function $f(\mathbf{X}, \mathcal{G}_t)$, which reliably distinguishes true anomalies from normal operational fluctuations while providing causal explanations for its decisions.

This formulation leads to three key requirements that address challenges M1–M3. *Adaptive causal discovery (R1)* requires tracking the evolution of $\mathcal{E}_t$ over time without relying on stationarity assumptions, thereby addressing causal drift. *Computational efficiency (R2)* necessitates a lightweight learning framework suitable for resource-constrained ICS edge environments. *Interpretable inference (R3)* entails that anomaly decisions be explainable through causal paths in $\mathcal{G}_t$ to support root cause diagnosis.

B. Research Motivation

The primary motivation for this work arises from a systematic limitation observed in current methods: their inability to fulfill requirements R1–R3 concurrently. As demonstrated by the analysis in Section I and illustrated in Fig. 1, this shortcoming remains a persistent challenge, despite advances reported in individual aspects of prior studies. This gap is further illustrated

TABLE I
COMPARATIVE ANALYSIS OF ANOMALY DETECTION METHODS AGAINST KEY REQUIREMENTS FOR DYNAMIC ICS (BOLD VALUES HIGHLIGHT RL-ACID AS THE PROPOSED METHOD)

Method (Reference)	R1	R2	R3	Limitation
Statistical Models [2] [31]–[33]	×	✓	×	M1, M3
Deep Learning [8], [24]	×	-	×	M1, M3
Graph Neural Networks [9], [25]	-	-	-	M1, M2, M3
Static Causal Discovery [5], [19]	×	×	✓	M1, M2
Dynamic Causal [3], [30]	-	×	✓	M1, M2
RL-Based Detection [15], [16]	✓	×	×	M2, M3
RL-ACID (Proposed)	✓	✓	✓	Solves M1, M2, M3

by the comparative analysis of existing methods presented in Table I.

Statistical methods assume stationarity and capture only correlations, failing to address the M1. Deep learning approaches operate as opaque models without causal reasoning, lacking M3. Graph neural networks typically rely on static graph structures, while static causal discovery methods cannot handle time-varying causality. Although dynamic causal models incorporate temporal dynamics, they suffer from M2 and linearity constraints. RL methods demonstrate adaptability but lack causal interpretability and exhibit high computational costs. The critical research gap identified through this analysis is the absence of methods that can simultaneously achieve adaptive causal discovery, computational efficiency, and interpretable reasoning. This limitation directly motivates the development of our framework to bridge these interdependent requirements.

C. Novelty

The novelty of the RL-ACID framework is defined not merely by enhanced accuracy, but by its integrated causal architecture that systematically resolves the three intertwined challenges (R1–R3) in a unified, adaptive manner.

Architectural integration: RL-ACID represents the first framework that integrates lightweight reinforcement learning-based causal search (CaRLite), time-frequency causal encoding, and causal clustering into an end-to-end pipeline. This holistic design consciously addresses adaptation, efficiency, and interpretability as interdependent objectives rather than optimizing them in isolation.

Technical innovations: At the component level, we introduce three key innovations. For adaptive causal discovery (R1), we develop a causal clustering module with structural experts and soft assignment that enables unsupervised identification and smooth switching between time-varying causal regimes. For computational efficiency (R2), we design structured sparse attention [lightweight industrial sparse attention (LISA)] and prior-constrained RL search (CaRLite) to achieve algorithmic-level optimizations suitable for edge deployment. For interpretable inference (R3), the framework centers on causal graphs as intermediate representations, incorporating causal path tracing to transition from anomaly detection to root-cause diagnosis.

The subsequent section details the methodological implementation of the RL-ACID framework, building upon these foundational contributions.

III. RL-BASED ANOMALY DETECTION FRAMEWORK FOR CAUSAL DISCOVERY

A. Data Coding and Feature Representation

In causal discovery tasks, high-quality time-series representations are crucial for ensuring the accuracy of structural searches. In ICS time series, it has been shown that combining time-domain and frequency-domain features enhances feature extraction [17]. To more effectively capture both temporal and spectral characteristics in multivariate industrial time series, we propose a time-frequency enhanced transformer encoder-decoder architecture as a unified framework for causal feature modeling and causal graph generation. By integrating a self-attention mechanism with directed acyclic graph (DAG) structural constraints, the architecture effectively reveals potential causal relationships among variables.

Encoder: Time-frequency causal feature extraction. To enhance the characterization of anomalous causal pathways, we introduce a hybrid encoder that integrates time-frequency analysis with temporal modeling. Raw multivariate sensor signals are segmented into local blocks $\mathbf{X}_t \in \mathbb{R}^{W \times D}$ using a sliding window. Each block is transformed into the frequency domain via fast Fourier transform, producing a complex-valued representation $\mathbf{F}(t) \in \mathbb{C}^{W \times D}$. The amplitude spectrum $|\mathbf{F}(t)|$ is normalized to improve robustness to temporal misalignments and capture global spectral structures. Such time-frequency representations have been shown to complement time-domain features by modeling long-range dependencies and dynamic couplings in nonstationary industrial processes [18]. The resulting spectral representation is fused with temporal features to form a hybrid encoding. This joint representation enhances multiscale feature characterization while preserving both temporal dynamics and spectral regularities, providing a stronger basis for identifying stable periodic causal relationships essential for downstream inference in complex industrial environments.

Subsequently, a multiscale frequency-domain convolutional module extracts discriminative spectral features, emphasizing oscillatory patterns and multivariable interactions through hierarchically stacked convolutional operations. The extracted

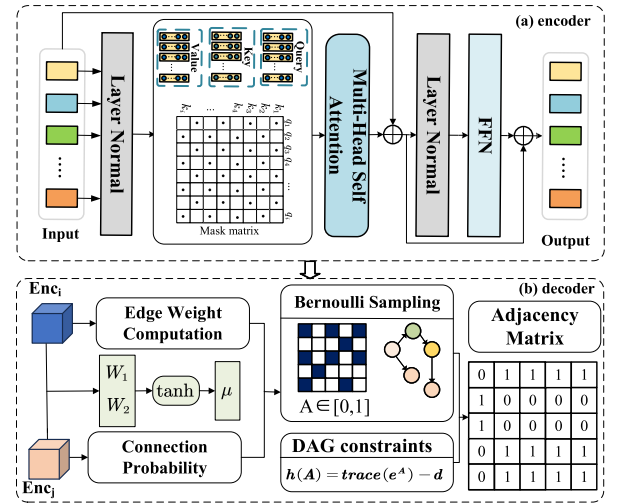


Fig. 3. Architecture of the encoder-decoder model for causal structure modeling.

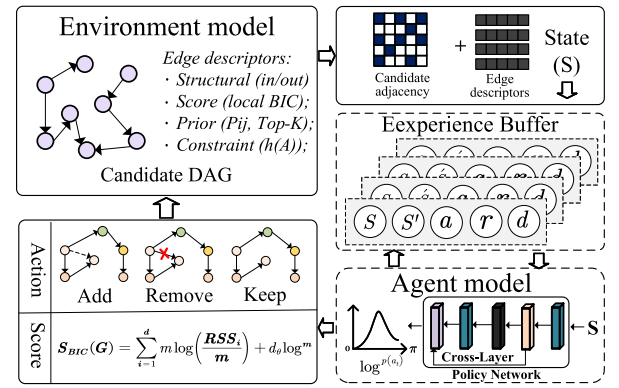


Fig. 4. Architecture for causal discovery with reinforcement learning-optimized dynamic structure learning.

spectral features are concatenated with the time-domain representations, yielding a unified time-frequency feature vector that provides causally informative input to the transformer.

LISA: Industrial monitoring data exhibit quasi-periodic patterns and response delays, with most causal influences confined to short temporal neighborhoods and only a few long-cycle effects at characteristic lags. Conventional self-attention captures these dependencies at high computational cost—scaling as $\mathcal{O}(W^2 D^2)$ for long windows or many sensors. To reduce complexity, LISA employs a *structurally guided sparse self-attention* mechanism [see Fig. 3(a)]. For each token (t, i) , a binary mask $\mathbf{M}$ restricts attention to (i) a local temporal band of radius r around t for short-term causal interactions and (ii) dilated sampling points with stride Δ for delayed and long-range periodic effects.

Formally, the input features $\mathbf{H} \in \mathbb{R}^{n \times d}$ are projected into query, key, and value matrices using learnable weight matrices $\mathbf{W}_Q, \mathbf{W}_K, \mathbf{W}_V$, where $\mathbf{W}_Q, \mathbf{W}_K \in \mathbb{R}^{d \times d_K}$ and $\mathbf{W}_V \in \mathbb{R}^{d \times d_V}$. The sparse self-attention for the i th head is then computed as

$$\text{Head}_i = \text{Softmax} \left(\frac{\mathbf{Q}_i \mathbf{K}_i^\top}{\sqrt{d_K}} + \log \mathbf{M}_i \right) \mathbf{V}_i \quad (1)$$

where $\log \mathbf{M}_i$ assigns 0 to valid entries and $-\infty$ to masked positions so excluded interactions are not included in the softmax.

With band radius r , dilated count K , and optional subsampling stride p , LISA ensures that each query attends to at most $B \approx (2r + K) + \lceil W/p \rceil$ keys—typically $B \ll WD$. This reduces the computational complexity from $\mathcal{O}(W^2D^2)$ to $\mathcal{O}(WDB)$ while retaining the ability to capture both immediate causal dynamics and delayed periodic effects that are characteristic of nonstationary industrial environments and critical for downstream causal discovery.

Decoder: Causal graph structure generation: The specialized decoder module [see Fig. 3(b)], built upon a time-frequency encoder for multivariate ICS, generates causal structures from the encoded representations. Unlike traditional decoders used for regression or classification tasks, this module generates directed edges directly between variables to construct a directed DAG that drives RL optimization. This design ensures a tight coupling between causal modeling and data representation. The decoder efficiently converts the joint time-frequency representation into causal relationships to form the initial causal graph $\mathcal{G} = (V, E)$. Although not based on traditional causal discovery algorithms, the graph represents plausible causal relationships because the edges encode directional dependencies inferred from time-frequency features. Subsequent optimization using the Bayesian information criterion (BIC) and linear parent-child modeling further supports interpreting this graph as a causal structure.

Given the encoder outputs $[\text{Enc}_1, \text{Enc}_2, \dots, \text{Enc}_n]$, which encode both temporal and frequency-domain dependencies across variables, the decoder estimates pairwise causal relations via a neural scoring mechanism. For any variable pair (i, j) , the edge existence probability is computed as

$$A_{ij} = \text{Bernoulli}(\sigma(\mathbf{u}^\top \tanh(W_1 \text{Enc}_i + W_2 \text{Enc}_j))) \quad (2)$$

where $\mathbf{W}_1, \mathbf{W}_2$ are learnable linear transformation matrices, $\mathbf{u}$ is the causality score vector, and $\sigma(\cdot)$ denotes the Sigmoid function. This process models the probability for each edge in the adjacency matrix $\mathbf{A}$. To enforce acyclicity, a smooth DAG constraint [19] is applied during training. After training, we retain the top- k edges with the highest probabilities and project them onto an acyclic graph. This produces a high-confidence coarse causal graph that not only confines the initial search space for the subsequent RL but also significantly enhances computational efficiency without compromising detection performance. This decoder also acts as the policy model in the RL framework, whose edge probabilities define a structured action space and are optimized via policy gradient methods to maximize a reward function combining causal plausibility and detection performance (see Section III-B).

B. RL-Based Causal Structure Search

We formulate causal discovery under dynamic industrial conditions as a sequential decision process. To tackle causal drift and computational complexity, the proposed RL framework is built around a structured environment, a sparsity-aware reward, and an experience replay system, which collectively enable an adaptive and efficient search (see Fig. 4 and Algorithm 1). The

Algorithm 1: CaRLite: Lightweight Causal Structure Search.

Input: Initial DAG G_0 from encoder-decoder;
candidate edge pool $\mathcal{E}_0$; local BIC oracle.

Output: Refined DAG $\hat{G}$

```

1 Initialize policy  $\pi_\theta$ , hierarchical buffer  $\mathcal{B}$ , temperature  $\lambda$ ;  $A_0 \leftarrow \text{Adj}(G_0)$ 
2 while not converged do
3   Construct state  $s_t \leftarrow (A_t, \{\phi_{ij}\})$ 
4   Sample edge action  $a_t \sim \pi_\theta(\cdot | s_t; \lambda)$ 
5   Update graph  $G_{t+1}$  by applying  $a_t$ , reject if cyclic
6   Compute reward  $r_t$  using BIC score and constraints
7   // Intelligent experience management Store
    $(s_t, s_{t+1}, a_t, r_t, d_t)$  in  $\mathcal{B}$  by stability tier
8   Sample from  $\mathcal{B}$  with causal drift awareness
9   Update policy via REINFORCE with baseline
10  Adjust  $\lambda$  for exploration-exploitation balance
11  if no improvement for  $L$  steps then
12     $d_t \leftarrow \text{True}$  break
13 Prune non-positive edges while maintaining acyclicity
14 return  $\hat{G}$ 

```

framework comprises three core components: an *environment model*, an *agent model*, and the causal RL-based structure search (CaRLite) algorithm. Integrating them via hierarchical experience management, CaRLite achieves a 68% gain in computational efficiency through intelligent reuse.

1) *Environment model:* The RL environment models the adaptive causal discovery process as a dynamic agent-environment interaction (Algorithm 1, line 3–6). Within this framework, the agent iteratively explores and refines candidate DAGs, receiving feedback after each structural change. To enable stable and efficient policy learning, the environment provides a well-structured state representation (Algorithm 1, line 3) and a reward mechanism (Algorithm 1, line 6) that guides the agent toward uncovering causal relationships that are both fine-grained and reflective of the current process.

State-space construction (Algorithm 1, line 3) integrates the candidate adjacency matrix with edge-level descriptors ϕ_{ij} , which combine structural features, score-based metrics, decoder priors, and DAG constraints. This design provides the agent with essential structural and statistical cues while keeping per-step computation tractable.

Reward mechanism (Algorithm 1, line 6): When the agent modifies the graph structure, the environment returns a composite reward that balances model fit (BIC score), sparsity (ℓ_0 -norm), and acyclicity (NOTEARS metric). The reward leverages BIC decomposability for efficient local rescoring, delivering immediate feedback while maintaining computational efficiency.

2) *Agent model:* The agent iteratively refines a DAG by selecting edge operations (addition, deletion, retention) based on the structured state (Algorithm 1, line 5). We employ a compact hierarchical policy network that preserves structural information while avoiding redundant computation, complemented by an adaptive temperature-decay mechanism (Algorithm 1, line 10)

that dynamically balances exploration and exploitation throughout training.

3) *CaRLite algorithm*: Algorithm 1 outlines the complete causal structure search procedure, implementing an efficient RL process for causal discovery. The algorithm accepts an initial DAG G_0 from the encoder-decoder module, a candidate edge pool $\mathcal{E}_0$, and a local BIC oracle, and returns a refined DAG $\hat{G}$ as output.

The procedure initializes the policy network π_θ , a hierarchical experience buffer $\mathcal{B}$, and the temperature parameter λ (line 1). The main search loop (line 2) iterates until convergence, with each iteration executing the following steps in sequence. First, a structured state representation is constructed by combining the current adjacency matrix with edge-level descriptors (line 3). Based on this state, the policy network samples an edge-modification action (line 4), which is applied to update the graph while enforcing acyclicity constraints (line 5). Following each structural change, a composite reward is computed using the BIC score and structural penalties (line 6). To manage past experiences effectively, an experience replay mechanism is employed (lines 7–8). This mechanism stores transitions in a tiered buffer according to their learning stability and retrieves them for policy updates based on the detected magnitude of causal drift. Subsequently, the policy parameters are updated using the REINFORCE algorithm with a baseline (line 9), while the temperature parameter λ is annealed to balance exploration and exploitation (line 10). Convergence is detected when no performance improvement occurs for L consecutive steps (lines 11–13), at which point the search terminates. Finally, nonpositive edges are pruned while preserving acyclicity, yielding the optimized DAG $\hat{G}$.

The policy maximizes the expected return under the composite reward $R(G) = -[S_{\text{BIC}}(G) + \Phi(G)]$. Here, $S_{\text{BIC}}(G)$ penalizes model complexity while assessing fit, given by

$$S_{\text{BIC}}(G) = \sum_{i=1}^d \left[m \log \left(\frac{\text{RSS}_i}{m} \right) \right] + d_\theta \log m \quad (3)$$

where $\text{RSS}_i = \sum_{k=1}^m (x_{ki} - \hat{x}_{ki})^2$ is the residual sum of squares for node i , and $d_\theta = \sum_{i=1}^d |\text{Pa}(i)|$ is the total number of parent nodes across the graph. The structural penalty $\Phi(G)$ is defined as

$$\Phi(G) = \lambda_1 \mathbf{I}(G \notin \text{DAGs}) + \lambda_2 h(A) + \mu \|A\|_0 \quad (4)$$

where $h(A) = \text{trace}(e^A) - d$ enforces the DAG constraint through a smooth acyclicity metric, and the ℓ_0 -norm $\|A\|_0$ promotes sparsity.

The policy network employs a lightweight structure comprising only 3–4 layers with cross-layer feature pathways, designed to preserve multiscale structural cues while maintaining efficiency. Training is stabilized by an adaptive temperature-decay mechanism λ , which ensures a smooth transition from exploration to exploitation as the policy converges.

This structured approach enables CaRLite to achieve a 68% gain in computational efficiency through intelligent experience reuse, while directly promoting the discovery of sparse, acyclic, and high-fidelity causal graphs for dynamic industrial environments. The resulting DAG $\hat{G}$ captures time-varying causal

relationships and is subsequently passed to downstream modules for causal clustering and anomaly detection.

C. Causal Clustering Module and Anomaly Detection

To address causal drift under varying system conditions, we introduce a distribution routing mechanism that encodes each causal graph into a low-dimensional structural embedding and softly assigns it to M latent causal experts. Each expert captures a distinct structural regime, and their outputs are weighted and fused to support robust inference.

Distribution router: Each RL-generated adjacency matrix $\hat{G} \in \{0, 1\}^{d \times d}$ is first flattened and encoded by a two-layer MLP to produce a structural embedding

$$\mathbf{z}_t = \text{MLP}(\text{Flatten}(\hat{G})) \in \mathbb{R}^d. \quad (5)$$

The resulting structural embedding $\mathbf{z}_t \in \mathbb{R}^d$ is projected to M latent structural experts using a lightweight two-layer gating network

$$\mathbf{H}_t = \mathbf{U}_2 \cdot \text{ReLU}(\mathbf{U}_1 \cdot \mathbf{z}_t) \in \mathbb{R}^M \quad (6)$$

where $\mathbf{U}_1 \in \mathbb{R}^{d \times d_h}$ and $\mathbf{U}_2 \in \mathbb{R}^{d_h \times M}$ are learnable parameters. Here, d is the dimension of the structural embedding, d_h is the hidden dimension of the gating network, and M denotes the number of structural experts.

The model then computes a sparse routing distribution over the top- K selected experts via a masked softmax, assigning a normalized weight $\mathbf{w}_t^{(i)}$ to each expert based on its compatibility with the current causal graph. This deterministic approach clusters time-varying structures without stochastic reparameterization, improving stability and interpretability in nonstationary environments.

Expert prediction and aggregation: Each expert $\mathcal{E}_i$ maintains a linear mechanism $f_i(\cdot)$ to predict the target variable x_j from its parent set pa_j

$$\hat{x}_j^{(i)} = \mathbf{w}_j^{(i)\top} \cdot \text{pa}_j + b_j^{(i)} \quad (7)$$

where $\mathbf{w}_j^{(i)} \in \mathbb{R}^{|\text{pa}_j|}$ is the expert-specific weight vector and $b_j^{(i)}$ is a learnable bias term.

The final prediction is the weighted sum of expert outputs

$$\hat{x}_j = \sum_{i=1}^K w_t^{(i)} \cdot \hat{x}_j^{(i)}. \quad (8)$$

This expert-driven routing and fusion strategy allows the model to adapt to time-varying causal structures, enhancing robustness, and interpretability under nonstationary dynamics.

After the RL search converges and yields the final causal graph $\hat{G}$, the anomaly detection module is trained separately. Given the discovered graph, the reconstruction network predicts future sensor values $\hat{X}_{t+1}^{(\hat{G})}$, and the prediction loss is defined as

$$\mathcal{L}_{\text{pred}} = \frac{1}{N} \sum_{t=1}^N \|\hat{X}_{t+1}^{(\hat{G})} - X_{t+1}\|_2^2. \quad (9)$$

This term measures the sum of squared prediction errors across all sensors between predicted and observed sensor readings and is used only to train the anomaly detection MLP and the causal clustering experts.

D. Theoretical Foundations and Analysis

To establish a rigorous mathematical foundation for RL-ACID, we formulate its core objective as a constrained multi-objective optimization problem that jointly learns time-varying causal structures $\{\mathcal{G}_t\}$ and the anomaly detection function f

$$\min_{\{\mathcal{G}_t\}, f} \mathbb{E}[\mathcal{L}_{\text{det}}] + \lambda_1 \|\mathcal{G}_t\|_0 + \lambda_2 R_{\text{DAG}}(\mathcal{G}_t) + \lambda_3 \Omega(f). \quad (10)$$

In this formulation, $\mathcal{L}_{\text{det}}$ quantifies the reconstruction error for detection accuracy, the ℓ_0 -norm enforces graph sparsity, R_{DAG} represents the NOTEARS constraint ensuring acyclicity, and $\Omega(f)$ controls model complexity. The coefficients $\lambda_1, \lambda_2, \lambda_3 \geq 0$ formally balance the inherent tradeoffs among these objectives, which correspond to the core challenges M1–M3. It is important to note that in our framework, these tradeoffs are intrinsically addressed through dedicated algorithmic designs rather than manual tuning of independent parameters.

This formulation provides the basis for three key theoretical innovations in RL-ACID. First, causal discovery is reformulated as a constrained RL problem, where an agent’s policy explores the space of DAGs by optimizing a reward derived from the negative of the cost terms in (10). This approach avoids the combinatorial explosion inherent in traditional score-based searches, directly addressing the efficiency challenge (M2). Second, the integration of the LISA sparse attention mechanism with causal graph learning induces a synergistic coregularization effect. The attention mechanism focuses on salient temporal dependencies to inform the structural prior, while the evolving causal graph provides feedback to refine attention patterns. This synergy enhances the model’s adaptability to temporal dynamics (M1). Third, the causal clustering module addresses nonstationarity by modeling the data as a mixture model of distinct operational regimes. This allows the model to identify invariant causal mechanisms, $P(Y|PA(Y))$, across regimes, providing a principled method to distinguish true anomalies from operational drift.

These theoretical guarantees collectively provide the analytical foundation for the empirical results in Section IV, confirming that RL-ACID’s performance stems from its principled design rather than empirical coincidences.

IV. EXPERIMENTS

A. Experiments Settings

Dataset: To evaluate the generalization and adaptability of our method across diverse industrial scenarios, we use five representative datasets spanning water treatment, electric power systems, and chemical processes: Secure Water Treatment (SWaT) [20], Water Distribution (WADI) [21], Hardware-in-the-Loop based Augmented Industrial Control System security (HIL-HAI) [4], the power system datasets (PSD) [22], and the Tennessee Eastman (TE) process benchmark [23]. These datasets vary in dimensionality, process complexity, control logic, and attack patterns. Table II summarizes the key characteristics of the datasets.

SWaT: A water treatment system with six stages and 51 variables, spanning 11 days (7 normal, 4 attack). It features clear phase transitions and control logic, making it ideal for analyzing causal drift and root cause chains.

TABLE II
DATASET CHARACTERISTICS AND OPERATIONAL COMPLEXITY

Dataset	Train	Test	Dimensions	Anomalies(%)	Stage
SWaT	496,800	449,919	51	11.98	6
WADI	1,048,571	172,801	123	5.99	10+
HIL-HAI	550,800	444,600	59	8.19	4
PSD	22,714	31,296	128	29.7	30+
TE	10,110	21,098	52	83.4	5

WADI: A large-scale water distribution system with 123 variables and 16 days of data (14 normal, 2 attack). Its higher dimensionality and dynamic conditions pose challenges for consistent modeling in multistage environments.

HIL-HAI: A hardware-in-the-loop ICS simulation with four coupled processes and 59 variables. It includes 10 days of normal data and 5.5 days of attack scenarios. Its frequent condition changes demand strong adaptability to structural variations. Version 1.0 is used.

PSD: Phasor measurement unit (PMU) traces from a benchmark power grid with 37 operating scenarios covering normal operation, natural disturbances and three types of cyber-physical attacks. For our study, these scenarios are grouped into normal and abnormal to perform binary anomaly detection.

TE: A highly nonlinear and tightly coupled chemical process simulation with 52 variables and 21 diverse fault types, including step changes, slow drifts, and valve failures. It provides a long-horizon and imbalanced benchmark for evaluating anomaly detection in complex industrial environments.

Baseline: To comprehensively evaluate the effectiveness of the proposed RL-ACID framework, we compare it with representative baselines from both anomaly detection and causal discovery. For anomaly detection, we include Deep Robust One-Class Classification (DROCC) [24], a semisupervised deep model that learns a compact decision boundary around normal samples; Graph Deviation Network (GDN) [25], a graph-neural-network-based detector modeling neighborhood dependencies among sensors; Graph-Augmented Normalizing Flow (GANF) [26], Multivariate Time series anomaly detection via dynamic Graph and entityaware normalizing Flow (MTGFlow) [27], and a Graph Mixture of Experts (Graph-MoE) [28], which represent state-of-the-art unsupervised graph-based anomaly detectors; and Anomaly Root Cause Analysis via Granger Causal Discovery (AERCA) [29], which integrates Granger causality with root-cause analysis for interpretable anomaly detection. For causal discovery, we further consider DYNOTEARS [30], a score-based dynamic extension of NOTEARS for learning time-lagged causal graphs; PCMCi+ [5], a constraint-based method that combines conditional independence tests with false-discovery-rate control for high-dimensional time series; causal forests [31], an ensemble approach for estimating heterogeneous causal effects without strong parametric assumptions; and transfer entropy [32], an information-theoretic metric that quantifies directional dependencies between time-series variables.

To ensure a fair comparison in anomaly detection, each of these methods is first applied to the training data to learn a causal

TABLE III

COMPARISON OF F1-SCORE, AVERAGE PRECISION (A-PR), AND AVERAGE ROC (A-ROC) ON FIVE BENCHMARK DATASETS (BOLD VALUES INDICATE THE PROPOSED RL-ACID METHOD AND HIGHLIGHT BEST OR SUPERIOR PERFORMANCE)

Method	SWaT			WADI			HIL-HAI			PSD			TE		
	F1	A-PR	A-ROC	F1	A-PR	A-ROC	F1	A-PR	A-ROC	F1	A-PR	A-ROC	F1	A-PR	A-ROC
Transfer Entropy	61.0 $\pm$ 2.5	57.0 $\pm$ 2.3	70.0 $\pm$ 2.6	45.0 $\pm$ 3.1	42.0 $\pm$ 3.0	67.0 $\pm$ 3.2	60.0 $\pm$ 2.4	56.0 $\pm$ 2.2	70.0 $\pm$ 2.5	50.0 $\pm$ 3.0	47.0 $\pm$ 2.9	70.0 $\pm$ 3.1	58.0 $\pm$ 3.2	55.0 $\pm$ 3.0	75.0 $\pm$ 3.3
Causal Forests	65.5 $\pm$ 2.2	61.2 $\pm$ 2.1	71.5 $\pm$ 2.3	48.0 $\pm$ 2.9	45.0 $\pm$ 2.8	70.0 $\pm$ 3.0	64.0 $\pm$ 2.3	60.0 $\pm$ 2.2	72.0 $\pm$ 2.4	54.0 $\pm$ 3.1	51.0 $\pm$ 2.9	73.0 $\pm$ 3.2	62.0 $\pm$ 3.0	59.0 $\pm$ 2.8	80.0 $\pm$ 3.1
PCMCi+	67.0 $\pm$ 2.3	63.0 $\pm$ 2.2	72.0 $\pm$ 2.4	52.0 $\pm$ 3.0	48.0 $\pm$ 2.9	73.0 $\pm$ 3.2	67.0 $\pm$ 2.2	63.0 $\pm$ 2.0	74.0 $\pm$ 2.3	57.0 $\pm$ 3.2	54.0 $\pm$ 3.0	77.0 $\pm$ 3.3	64.0 $\pm$ 3.1	61.0 $\pm$ 2.9	83.0 $\pm$ 3.2
DYNOTEARS	70.0 $\pm$ 2.1	66.0 $\pm$ 2.0	74.0 $\pm$ 2.2	54.0 $\pm$ 2.8	50.0 $\pm$ 2.7	76.0 $\pm$ 2.9	68.0 $\pm$ 2.0	65.0 $\pm$ 1.9	76.0 $\pm$ 2.1	59.0 $\pm$ 2.9	56.0 $\pm$ 2.7	80.0 $\pm$ 3.0	66.0 $\pm$ 2.8	63.0 $\pm$ 2.6	85.0 $\pm$ 2.9
DROCC	72.3 $\pm$ 2.9	68.7 $\pm$ 2.5	72.6 $\pm$ 3.8	57.4 $\pm$ 2.1	54.0 $\pm$ 2.0	75.6 $\pm$ 1.6	70.2 $\pm$ 2.6	67.0 $\pm$ 2.3	74.4 $\pm$ 2.8	60.5 $\pm$ 1.6	58.2 $\pm$ 1.5	78.0 $\pm$ 1.7	65.0 $\pm$ 1.8	62.5 $\pm$ 1.6	82.0 $\pm$ 1.9
GDN	77.6 $\pm$ 2.4	75.3 $\pm$ 2.0	78.5 $\pm$ 2.1	62.2 $\pm$ 2.2	60.0 $\pm$ 1.9	82.7 $\pm$ 1.4	76.5 $\pm$ 2.1	74.2 $\pm$ 1.9	79.5 $\pm$ 1.8	66.8 $\pm$ 1.5	64.1 $\pm$ 1.4	84.5 $\pm$ 1.5	70.5 $\pm$ 1.7	67.9 $\pm$ 1.6	87.2 $\pm$ 1.6
GANF	78.5 $\pm$ 1.8	76.3 $\pm$ 1.5	79.8 $\pm$ 0.7	64.2 $\pm$ 1.4	62.0 $\pm$ 1.1	90.3 $\pm$ 1.0	77.8 $\pm$ 1.3	75.5 $\pm$ 1.2	80.6 $\pm$ 1.2	68.9 $\pm$ 1.4	66.0 $\pm$ 1.3	86.0 $\pm$ 1.4	72.0 $\pm$ 1.5	69.5 $\pm$ 1.3	88.8 $\pm$ 1.3
MTGFlow	81.1 $\pm$ 1.6	78.4 $\pm$ 1.5	84.8 $\pm$ 1.5	66.2 $\pm$ 1.3	64.0 $\pm$ 1.1	91.9 $\pm$ 1.1	81.4 $\pm$ 1.2	79.0 $\pm$ 1.1	85.6 $\pm$ 1.4	71.5 $\pm$ 1.3	68.7 $\pm$ 1.2	89.2 $\pm$ 1.2	74.3 $\pm$ 1.2	71.0 $\pm$ 1.1	90.6 $\pm$ 1.1
Graph-MoE	83.5 $\pm$ 1.3	81.2 $\pm$ 1.2	87.2 $\pm$ 1.3	68.3 $\pm$ 1.1	66.1 $\pm$ 0.8	94.2 $\pm$ 0.8	82.7 $\pm$ 1.1	80.5 $\pm$ 1.0	88.5 $\pm$ 1.2	73.8 $\pm$ 1.1	71.5 $\pm$ 1.0	91.0 $\pm$ 1.0	78.0 $\pm$ 1.0	74.2 $\pm$ 0.9	92.5 $\pm$ 0.9
AERCA	82.1 $\pm$ 1.4	80.0 $\pm$ 1.3	86.4 $\pm$ 1.4	67.0 $\pm$ 1.2	65.0 $\pm$ 1.0	93.5 $\pm$ 1.0	80.3 $\pm$ 1.2	78.2 $\pm$ 1.1	87.5 $\pm$ 1.3	72.6 $\pm$ 1.2	70.2 $\pm$ 1.1	90.5 $\pm$ 1.1	75.0 $\pm$ 1.1	72.1 $\pm$ 1.0	91.8 $\pm$ 1.0
RL-ACID (Ours)	85.0 $\pm$ 1.2	82.1 $\pm$ 1.0	88.8 $\pm$ 1.3	68.1 $\pm$ 0.9	67.1 $\pm$ 0.7	94.3 $\pm$ 0.8	83.2 $\pm$ 1.1	80.5 $\pm$ 0.9	90.2 $\pm$ 1.2	75.9 $\pm$ 1.0	73.0 $\pm$ 0.9	92.7 $\pm$ 0.9	78.2 $\pm$ 1.0	74.9 $\pm$ 0.9	92.3 $\pm$ 0.9

graph, which is then fed into the same downstream anomaly-detection module with a unified thresholding scheme. This set of baselines enables a balanced evaluation of RL-ACID against both leading anomaly detection techniques and representative causal structure learning methods.

Evaluation metrics: Following previous works [27], [28], we adopt a window-level evaluation strategy, where a window is labeled as anomalous if any time point within it is detected as abnormal. The primary evaluation metric is the area under the receiver operating characteristic curve (A-ROC), which reflects the tradeoff between true positive rate and false positive rate under varying thresholds. In addition to A-ROC, we report F1-score and area under the precision–recall curve (A-PR) in the main comparison experiments. F1-score captures the balance between precision and recall, while A-PR highlights performance on rare anomalies, which is particularly informative for imbalanced datasets. For F1 and A-PR, we determine the anomaly threshold by selecting the value that maximizes the F1-score on the validation set. This multimetric evaluation provides a comprehensive and robust assessment of detection accuracy across diverse conditions.

Implementation details: To ensure consistency and comparability, we adopt a unified experimental setup across all five datasets. The sliding window size and stride are set to 60 and 10, respectively, for all datasets. Model training employs the Adam optimizer with dataset-specific hyperparameters: for SWaT and HIL-HAI (medium-scale systems), we use a batch size of 512 and learning rate of 0.005; for WADI and PSD (large-scale systems), the batch size is 320 with a learning rate of 0.0035; and for TE (highly imbalanced system), we use a batch size of 256 and learning rate of 0.002. The causal graph construction retains a fixed number of high-confidence edges to maintain structural sparsity across all datasets. All experiments are conducted on a server equipped with NVIDIA RTX 3090 GPUs and 64 GB RAM. To ensure statistical reliability, the results of our proposed model are calculated as mean values from 5 independent runs with different random seeds.

B. Main Comparison and Analysis.

We begin by presenting in Table III, a summary of the RL-ACID’s performance against baselines across five ICS datasets. RL-ACID demonstrates the best overall anomaly detection performance on the medium-scale ICS testbeds SWaT and HIL-HAI. It attains the highest scores on SWaT, with 85.0% in F1, 82.1% in A-PR, and 88.8% in A-ROC. Similarly, on HIL-HAI, it reaches 83.2% in F1, 80.5% in A-PR, and 90.2% in A-ROC, consistently outperforming all baseline methods. In contrast, static causal models like PCMCi+ and DYNOTEARS yield lower F1 and A-PR performance. This advantage stems from RL-ACID’s integrated framework: a time-frequency encoder that captures underlying system dynamics, combined with RL-driven graph optimization that continuously adapts to operational variations—a capability static models lack.

The WADI dataset presents a challenging environment due to its high dimensionality (123 variables) and frequent operational shifts. On this dataset, RL-ACID achieves the best A-PR (67.1%) and A-ROC (94.3%). These results demonstrate its superior ranking capability and effectiveness in handling imbalanced anomaly detection tasks. Notably, the causal clustering module contributes significantly to this performance. It enhances model robustness by accurately separating true anomalies from normal operational variations, a capability where static models often fail. Although the F1-score of RL-ACID (68.1%) is slightly lower than that of Graph-MoE, this tradeoff underscores our focus on maintaining robust detection performance across varying thresholds rather than optimizing for a single operating point.

PSD is characterized by high dimensionality (128 variables) and diverse event scenarios (37 in total). RL-ACID achieves the strongest detection (75.9% F1, 73.0% A-PR, 92.7% A-ROC). The high variance and noise typical of PMU data make this dataset particularly challenging for classic causal methods, which show sharp performance drops. By contrast, RL-ACID’s time-frequency features combined with RL-based graph search improve robustness to noisy measurements and yield stable anomaly detection across operating regimes.

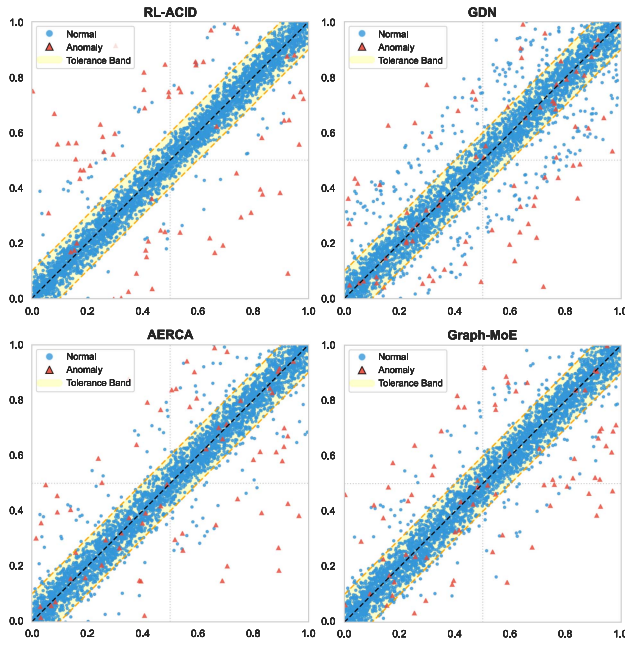


Fig. 5. Prediction-error scatter plots showing model sensitivity. Points within the yellow band have errors below the 0.2 tolerance. Red points are true anomalies: those outside the band are detected, inside are missed. Normal points outside the band represent false positives.

The TE process is a nonlinear, strongly coupled chemical system with 52 variables and 21 injected faults, exhibiting high imbalance (anomaly ratio $> 80\%$). RL-ACID achieves the best F1 (78.2%) and A-PR (74.9%), though its A-ROC (92.3%) is slightly below Graph-MoE. This is expected given the diversity of TE faults (valve failures, slow drifts, random variations), where some subtle anomalies are more readily captured by correlation-based ensembles. Nevertheless, the RL-ACID's superior performance on this strongly nonlinear system validates the effectiveness of our approach, where nonlinear feature extraction complements the linear BIC scoring to handle complex industrial dynamics.

Overall trends: Classical causal discovery baselines (transfer entropy, causal forests, PCMCi+, DYNOTEARS) show lower and more volatile performance, particularly on high-dimensional or regime-varying datasets (WADI, PSD, TE), because they assume either static structures or linear relationships. Modern graph-based baselines (Graph-MoE, AERCA) perform competitively but still lack adaptability to causal drift. The consistent improvements achieved by RL-ACID across both threshold-dependent (F1) and threshold-free (A-PR, A-ROC) evaluations confirm its superiority and robustness in dynamic ICS environments.

C. Qualitative Analysis and Interpretability

To evaluate anomaly detection accuracy and prediction stability, we compare the prediction performance of different models from the perspective of error structure, as shown in Fig. 5. Specifically, we plot the scatter of predicted versus observed values, and visualize the pointwise residuals to reflect the model's sensitivity to anomalies and its ability to control prediction errors. A yellow tolerance band is used to indicate

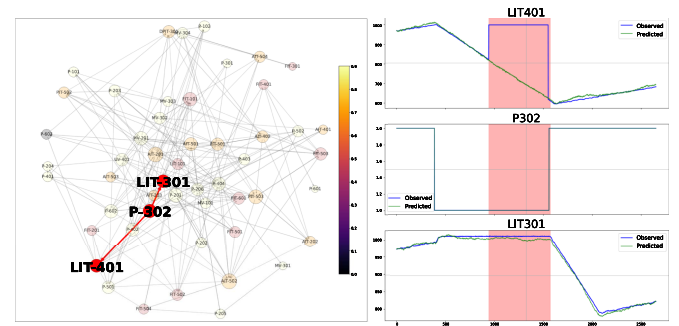


Fig. 6. Abnormal propagation effect under the perception of causal chains.

acceptable deviation, where points outside the band typically correspond to anomalies. Experimental results show that the proposed RL-ACID method produces more stable predictions under causal constraints, with normal samples densely clustered within the tolerance region and anomalies clearly located outside it, demonstrating strong discrimination capability. In contrast, alternative baselines yield more scattered residuals, resulting in higher false positive and false negative rates. This highlights the superiority of our approach in producing explainable and robust anomaly detection outcomes.

RL-ACID effectively supports anomaly localization and interpretation. As shown in Fig. 6 (left), the causal topology graph encodes degree-based node importance, functional-type color intensity, and edge thickness proportional to inferred causal strength. Without any prior structural assumptions, RL-ACID identifies meaningful causal dependencies, such as the chain $LIT401 \rightarrow P302 \rightarrow LIT301$, which aligns with known SWaT logic: LIT401 monitors the water level in a tank, which influences the operation of P302 (a pump), subsequently affecting LIT301, the downstream level sensor.

To evaluate interpretability under attack, we consider a scenario where LIT401's reading is maliciously locked at a high value. Since RL-ACID has captured LIT401's causal influence on P302, this manipulation causes P302 to remain open, unnecessarily pumping water and raising LIT301's level despite the tank being full. As shown in Fig. 6 (right), predictions match observations under normal operation but deviate significantly after the attack: the model expects P302 to close, yet it remains open, leading to unexpected rises in LIT301. This propagation of deviations along the causal chain demonstrates RL-ACID's capability to not only detect anomalies but also provide clear interpretive insights into their root causes.

To further validate the causal reasoning capability of RL-ACID, we systematically visualized the distribution of its reconstruction errors under multiple attack scenarios, as shown in Fig. 7. The heatmaps reveal distinct spatiotemporal error propagation patterns that closely correspond to specific attack types: stealth attacks exhibit a slow, progressive accumulation of errors; rapid cascading failures show sudden error bursts across critical nodes; and oscillatory attacks display periodic error peaks. Our comparison with the official SWaT attack documentation confirms that these residual distributions strongly align with the documented attack mechanisms. This correspondence demonstrates RL-ACID's ability to capture underlying causal

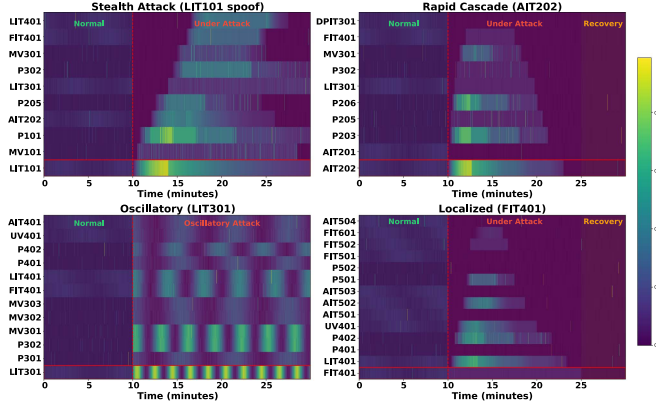


Fig. 7. Reconstruction error heatmaps for four representative SWaT attack scenarios: stealthy, rapid cascade, oscillatory, and localized.

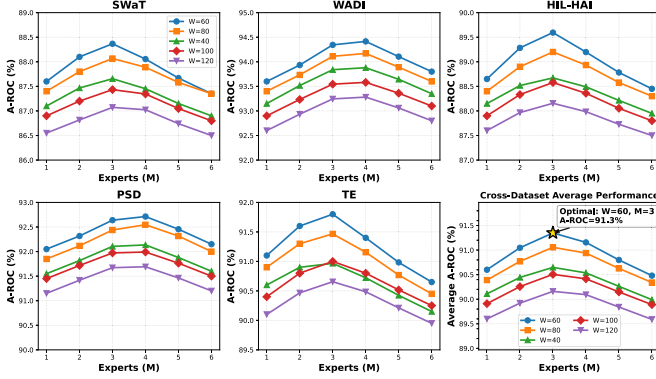


Fig. 8. Hyperparameter sensitivity analysis: A-ROC versus M and W .

dynamics without prior system topology knowledge. By reconstructing attack propagation paths through causal error analysis, our framework provides technicians with reliable diagnostic evidence for root cause identification and system recovery.

D. Hyperparameter Sensitivity Analysis

To evaluate the robustness of RL-ACID under different hyperparameter settings, we conduct sensitivity experiments on two key parameters: the sliding window size (W) and the number of experts (M) in the causal clustering module, as illustrated in Fig. 8. The results show that performance generally improves as W increases to around 60, beyond which overly large windows may dilute local anomaly patterns, while smaller ones lack sufficient temporal context. Regarding the number of experts, smaller-scale systems (SWaT, HIL-HAI) achieve their best performance at $M = 3$, whereas larger and more heterogeneous systems (WADI, PSD) prefer $M = 4-5$, with TE stabilizing around $M = 2-3$. The aggregated trend in Fig. 8 (bottom-right) shows a smooth and consistent response surface, indicating that RL-ACID is not overly sensitive to moderate variations of these parameters. In practice, W can be set according to the process cycle or sampling horizon, while M can be tuned based on the diversity of operating regimes, providing a systematic and interpretable guideline for deployment.

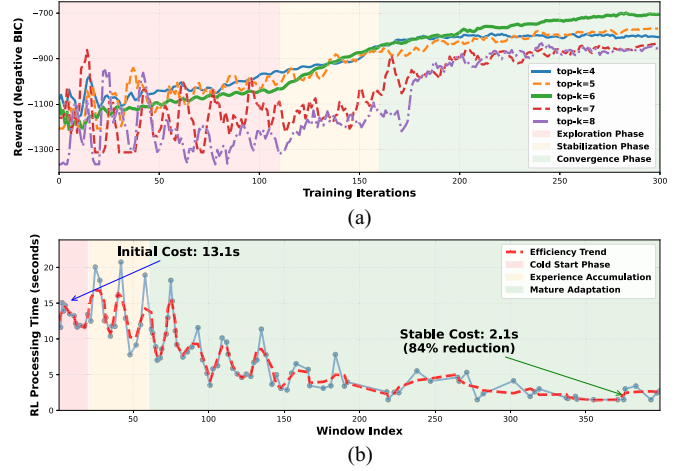


Fig. 9. (a) Analysis of policy convergence under different Top-K Values. (b) CaRLite computational cost over training.

TABLE IV
ABLATION RESULTS (A-ROC, %) ACROSS FIVE ICS DATASETS

Method	SWaT	WADI	HIL-HAI	PSD	TE
Full RL-ACID	88.8 $\pm$ 1.3	94.3 $\pm$ 0.8	90.2 $\pm$ 1.2	92.7 $\pm$ 0.9	92.3 $\pm$ 0.9
w/ Bi-GRU Encoder	86.9 $\pm$ 1.4	92.3 $\pm$ 0.9	88.4 $\pm$ 1.3	90.7 $\pm$ 1.0	90.4 $\pm$ 1.0
w/o RL Search	85.2 $\pm$ 1.6	91.0 $\pm$ 1.1	86.7 $\pm$ 1.5	89.0 $\pm$ 1.1	88.8 $\pm$ 1.1
w/o Causal Clustering	84.5 $\pm$ 1.6	90.3 $\pm$ 1.2	86.0 $\pm$ 1.5	88.4 $\pm$ 1.1	88.2 $\pm$ 1.1
w/o RL and CCM	81.5 $\pm$ 1.7	88.0 $\pm$ 1.4	83.0 $\pm$ 1.6	85.6 $\pm$ 1.3	85.0 $\pm$ 1.3

Results are mean $\pm$ Std over five runs.

Fig. 9 illustrates the dual optimization mechanisms of the RL-ACID framework. The upper panel demonstrates the trade-off between convergence speed and final performance under different Top-K pruning strategies: smaller K values (e.g., $K = 4$) achieve rapid convergence but suffer from performance limitations, while larger K values (e.g., $K = 8$) exhibit slow convergence and tend to settle at suboptimal solutions. $K = 6$ strikes an optimal balance, reaching peak performance within 120 iterations, validating the effectiveness of prior knowledge guidance. The lower panel showcases our innovative solution to the stability-plasticity dilemma through the CaRLite module. By integrating hierarchical experience replay with causal drift detection, we achieve intelligent balancing between experience reuse and adaptive exploration. Computational costs follow an exponential decay trend (68% overall reduction), while exploratory peaks triggered by causal structure changes prevent mode collapse in traditional RL and overcome the adaptability limitations of static models. This provides a lightweight yet robust adaptive causal discovery solution for dynamic industrial environments.

E. Ablation Studies

We further perform ablation studies across all five datasets to evaluate the contribution of each module in RL-ACID. As shown in Table IV, replacing the transformer encoder with a Bi-GRU leads to consistent but moderate drops (about 2%–3%

TABLE V
COMPUTATIONAL EFFICIENCY COMPARISON ON HIL-HAI AND WADI DATASETS (BOLD VALUES INDICATE THE PROPOSED RL-ACID METHOD AND HIGHLIGHT BEST OR SUPERIOR PERFORMANCE)

Method	Inference Time (ms)	Memory (MB)	Params (M)
DROCC	28.5 / 45.2	420 / 680	1.8
GDN	25.3 / 68.7	380 / 720	1.5
GANF	38.9 / 132.8	650 / 1650	3.2
MTGFlow	45.2 / 156.3	720 / 1980	3.8
Graph-MoE	58.7 / 215.6	890 / 2450	5.1
AERCA	67.3 / 245.6	920 / 2850	4.7
RL-ACID (w/o LISA)	68.9 / 198.4	1250 / 2850	4.2
RL-ACID (w/ global RL)	45.8 / 135.2	720 / 1300	4.2
RL-ACID (full)	42.5 / 128.3	680 / 1250	4.2

AUROC across datasets), confirming the benefit of attention-based encoding for capturing complex dependencies. Removing the RL-based search causes more severe declines (3%–4% on SWaT, HIL-HAI, and PSD), highlighting the importance of adaptive causal structure optimization. The causal clustering module is particularly critical on datasets with frequent operational shifts (−4.0% on WADI, −3.8% on TE), showing its ability to track structural drift under varying regimes. When both RL and clustering are removed, the degradation becomes sharp (over −6% on average, up to −7.5% on SWaT), underscoring their complementary effects. Overall, the ablation confirms that each component contributes substantially, and their combination of attention-based encoding, RL driven structure refinement, and causal clustering is essential for achieving robust and generalizable anomaly detection across diverse ICS environments.

F. Efficiency and Robustness Analysis

Computational efficiency and lightweight design are critical for deploying anomaly detection models in dynamic industrial environments. Table V provides a systematic evaluation on the HIL-HAI and WADI datasets. The complete RL-ACID framework achieves inference times of 42.5 ms (HIL-HAI) and 128.3 ms (WADI) with memory footprints of 680 MB and 1250 MB, respectively. This demonstrates a dual advantage: RL-ACID significantly outperforms complex ensemble methods (Graph-MoE, AERCA) in efficiency, while maintaining greater model capacity than simpler baselines (GDN).

RL-ACID's efficiency gains stem from its codesigned lightweight components, which are quantitatively validated through ablation studies. Removing the LISA sparse attention mechanism increases inference latency by approximately 62% and memory usage by 128% on the WADI dataset, confirming its essential role in alleviating quadratic complexity. Similarly, replacing the hierarchical experience replay in CaRLite with a global (nonhierarchical) search strategy leads to prolonged training and suboptimal convergence, despite comparable inference time. These results demonstrate that the LISA mechanism and the CaRLite algorithm are directly responsible for achieving a superior accuracy-efficiency tradeoff.

This lightweight design facilitates practical deployment. The RL agent operates exclusively during offline training, resulting in a stable model for online detection. With 4.2 million parameters and consistent sub-150 ms inference latency, RL-ACID

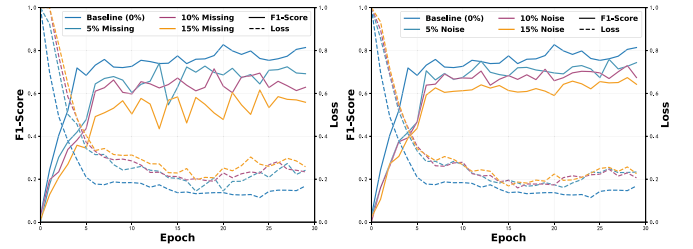


Fig. 10. Robustness analysis under data imperfections. Left: Performance degradation with increasing missing data ratios. Right: Training stability under varying noise levels.

demonstrates that principled, adaptive causal discovery can be achieved within the resource constraints of real industrial systems, effectively balancing high detection accuracy with operational efficiency.

To assess the practical viability of RL-ACID under realistic industrial conditions, we systematically evaluate its robustness against common data imperfections using the SWaT dataset. As illustrated in Fig. 10, we simulate two prevalent challenges: (left) varying ratios of missing data, and (right) increasing levels of additive Gaussian noise. Results indicate that although performance degrades with deteriorating data quality, the core functionality remains stable, demonstrating inherent resilience. This inherent resilience is attributed to the model's time-frequency feature encoding, which captures redundant spectral patterns, and the causal clustering module, which distinguishes structural drift from noise-induced perturbations. These findings confirm that RL-ACID is not only effective under idealized laboratory conditions but also retains robust functionality when confronted with the incomplete and noisy data streams typical of real-world industrial deployments.

V. CONCLUSION

This article has presented RL-ACID, an RL-optimized framework that addresses the challenge of adaptive causal discovery for robust anomaly detection in industrial systems. The methodological contribution lies in reformulating causal discovery as a constrained RL problem, which enables efficient exploration of graph structures while preserving interpretability. By integrating time-frequency feature encoding with a lightweight RL-based causal search and causal clustering, the framework effectively tracks time-varying causal relationships and distinguishes true anomalies from operational fluctuations. Extensive evaluations across five industrial benchmarks demonstrate RL-ACID's comprehensive advantages in detection accuracy, interpretability, and computational efficiency compared to existing baseline methods, establishing its superior performance in dynamic industrial environments.

Building upon the RL-ACID framework, future research will pursue two pivotal directions to broaden its industrial impact. First, we will focus on devising online and continual learning mechanisms for real time, incremental adaptation to streaming data. Second, we aim to establish federated causal discovery protocols that enable privacy-preserving, collaborative learning across distributed industrial sites. These research avenues are

designed to enhance the deployability, scalability, and practical utility of fully adaptive, data-driven causal anomaly detection in continuously evolving ICS.

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